COMP7640 E-COMMERCE PLATFORM

Group Member

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ER Diagram, Table Designs, and Normalization

ERDiagram

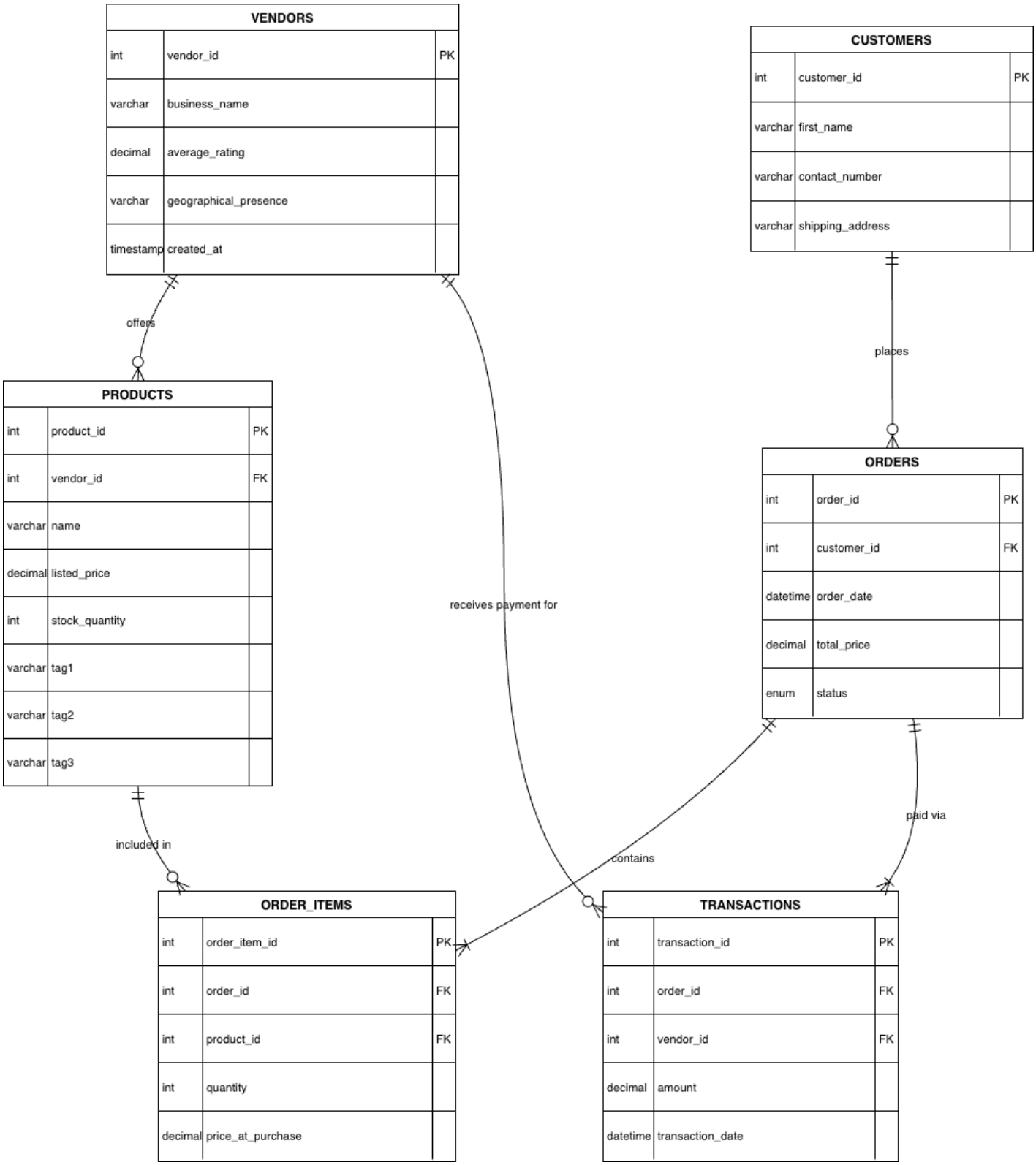


Table Designs

- **Vendors** Table: Uses `vendor_id` as the Primary Key. The `business_name` is set to `UNIQUE` to prevent duplicate registrations. A `CHECK` constraint ensures the `average_rating` stays between 0.00 and 5.00.
- **Products Table**: Linked to `vendors` via `vendor_id` (Foreign Key) with `ON DELETE CASCADE` to ensure that if a vendor is removed, their inventory is also deleted. It supports up to three tags (`tag1`, `tag2`, `tag3`) for product discovery.
- **Customers Table**: Stores buyer identity. `customer_id` is the Primary Key.
- **Orders Table**: Tracks order status using an `ENUM` type (`PENDING`, `PROCESSING`, `SHIPPED`, `DELIVERED`, `CANCELLED`). The `total_price` is updated based on the items purchased.
- **Order_Items Table**: Captures the state of a product at the time of purchase. It includes `price_at_purchase` to preserve historical financial records regardless of future price changes in the `products` table.
- **Transactions Table**: Implements the requirement for recording customer-to-vendor payments. It ensures that the `amount` paid to each vendor is correctly tracked for every order.

Normalization and Design Considerations

- **1NF**: all attrs are atomic value. we forbid multi value attr by using separate row for order item and transaction
- **2NF**: Each table has it own primary key, and all non key attrs are fully functionally dependent on the Primary key.
- **3NF**: we make sure there are no transitive dependency. ex. customer shipping detail are store in customer table rather than order to avoid redundancy.

Data Integrity & Robust

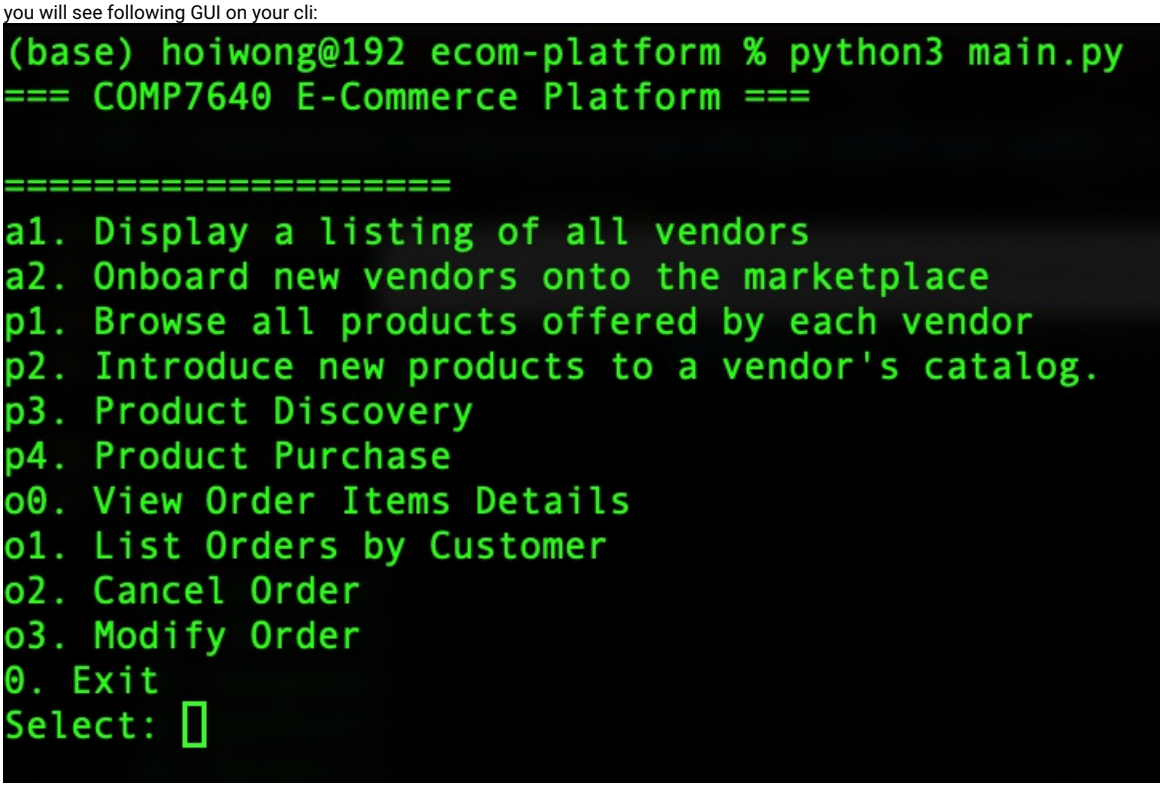
- **Check Constraints**: Applied to `listed_price`, `stock_quantity`, and `total_price` to prevent negative values, ensuring data reliability.
- **Referential Integrity**: Used `ON DELETE RESTRICT` for customer and product deletions in active orders to prevent orphaned records and maintain transaction history.

README for run code

prepare code

make sure ur python version is above 3.10.

```
# download code from github
git clone git@github.com:ava1anch33/ecom-platform.git
# go to the file
cd ecom-platform
# download all needed package
pip3 install -r requirements.txt
# check package are already download
pip3 list
# run project
python3 main.py
```

result

you will see following GUI on your cli: